

Tilak Bhusal

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Education

West Virginia University , Morgantown, USA	2024 – 2025
M.S., Industrial Engineering	
Tribhuvan University , Pulchowk, Nepal	2018 – 2023
B.S., Mechanical Engineering	

Professional Experience

Engineer	Energy Sciences	<i>Jan 2026 – Present</i>
• Spearhead comprehensive on-site facility assessments to identify high-impact energy efficiency and decarbonization opportunities across complex systems, including compressed air networks, VFDs, packaged RTUs, AHUs, and heat pumps. • Navigate state-specific Technical Reference Manuals (TRMs), regulatory codes, and emerging industry research to definitively quantify and validate custom energy savings. • Engineer customized calculation methodologies and build robust, industry-standard calculators to perform rigorous thermodynamic and economic analyses for energy efficiency measures. • Calculate and validate financial incentives for industrial and commercial clients, acting as a key technical liaison for utility-sponsored business energy efficiency programs. • Execute advanced Measurement and Verification (M&V) protocols to validate the performance of implemented projects, ensure sustained energy savings, and drive corrective actions. • Translate complex diagnostic data into actionable strategies by authoring detailed engineering reports and delivering compelling technical presentations to clients and cross-functional teams. • Author and optimize standard operating procedures (SOPs) and work instructions to ensure rigorous quality control and consistent engineering deliverables across all project phases.		
Graduate Research Assistant	WVU Pollution Prevention Lab	<i>Jan 2024 – Dec 2025</i>
• Conducted over 10 comprehensive energy assessments across various industrial sectors in West Virginia as part of ASHRAE Level I, II, and III audits. • Identified and quantified energy efficiency and pollution prevention opportunities in lighting, HVAC systems, motors, compressed air, steam, and process heating systems. • Utilized tools like MEASUR, eQUEST, and Excel-based calculators for energy modeling, equipment analysis, and ROI estimation.		
Intern	SunFarmer Nepal	<i>3/2022-8/2022</i>
• Designed solar arrays, pumps, and motors, and created CAD drawings for solar-powered projects. • Assisted in system optimization, project planning, and implementing renewable energy solutions for rural and underserved communities. • Conducted site monitoring and inspection, assisting in business development and procurement.		

Skills

Energy Auditing & Decarbonization: Energy Audits (ASHRAE Level I-III), Utility Bill Analysis, Measurement & Verification (M&V), Decarbonization Strategies, Equipment Diagnostics
Energy Modeling & Data Analytics: EnergyPlus, MEASUR, Python, R, MATLAB, Regression Modeling, Quantitative Analysis, Statistical Software (Jasp, Jamovi)
Mechanical Simulation & Design: ANSYS (CFD, FEA), SolidWorks (CAD)
Technical Compliance & Reporting: Technical Report Writing, Utility Incentive Calculation, Custom Calculator Development, Engineering Standard Operating Procedures (SOPs)

Certifications

- Certified Energy Manager (AEE)
- LEED Green Associate (GBCI)
- Energy Auditor Certification (Center for Energy Studies)
- Engineer-in-Training (NCEES)
- Registered Engineer, Nepal Engineering Council

Professional Development

- Member, Alpha Pi Mu (*Industrial Engineering Honor Society*)
- Association of Energy Engineers(AEE), Member